

Dexterous Manipulation by Two Fingers with Coupled Joints

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Overview

Future Work

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- ❖ Dynamics-based in-hand manipulation
- ❖ Contact modeling for fingers with *coupled* joints
- ❖ Object relocation and reorienting via torque control

- ❖ Experimental validation with Barrett Hand
- ❖ Grasping and reorienting with a 3D hand
- ❖ Kitchen knife pickup and rotation

Problem Description

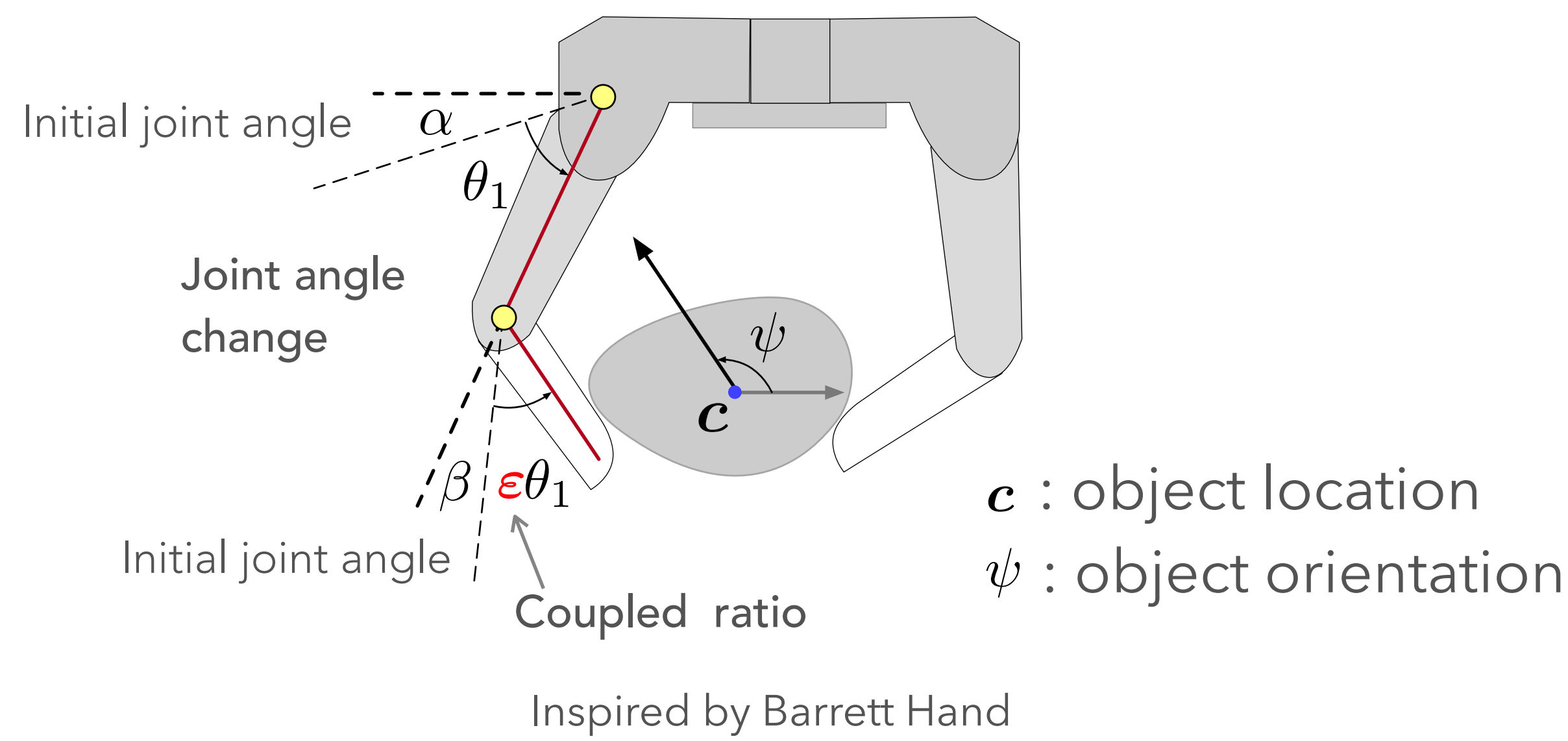
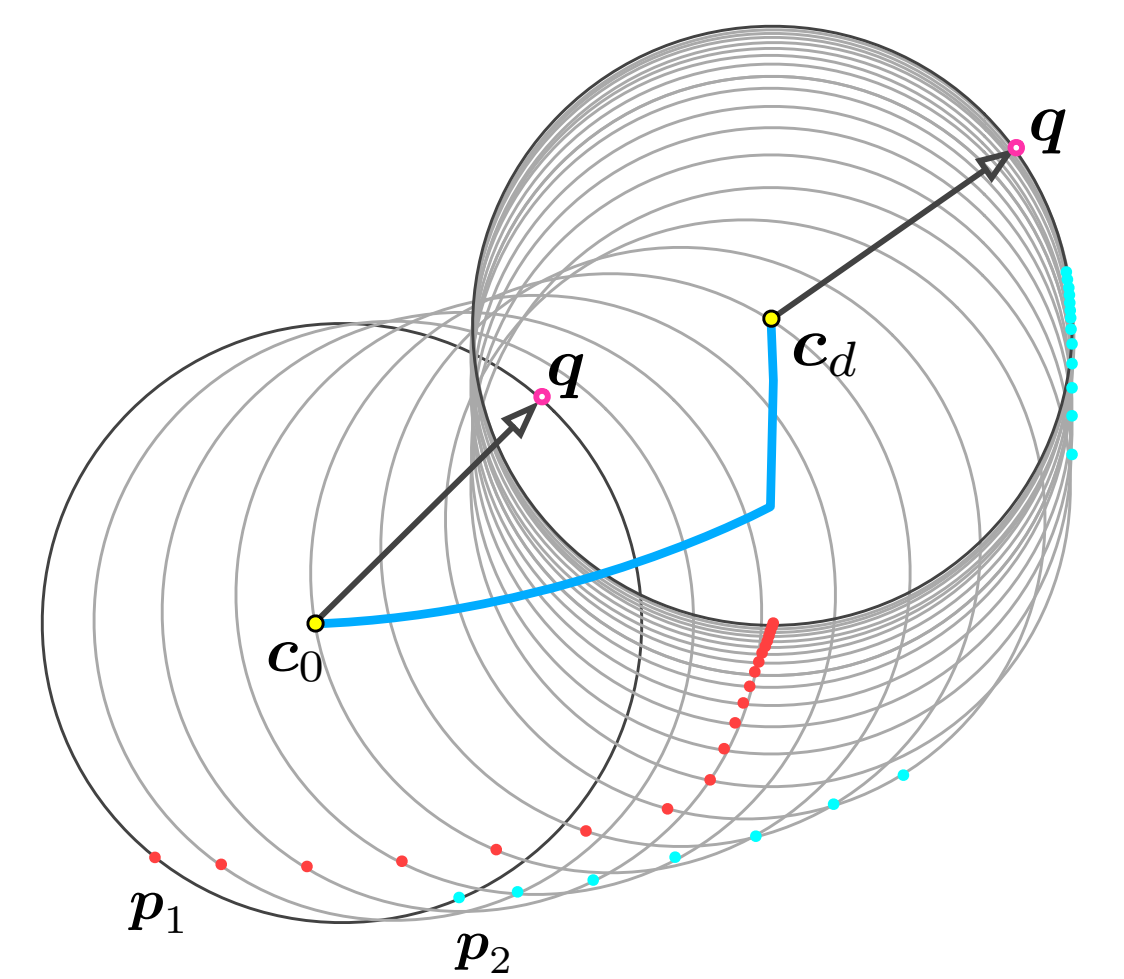
Object relocation and reorienting:

$$\mathcal{S}_0 = (c_0, \psi_0) \longrightarrow \mathcal{S}_d = (c_d, \psi_d)$$

Initial pose Goal pose

$$\zeta = (\theta_1, \theta_2, \psi)$$

$\zeta, \dot{\zeta}$: **state** of the system



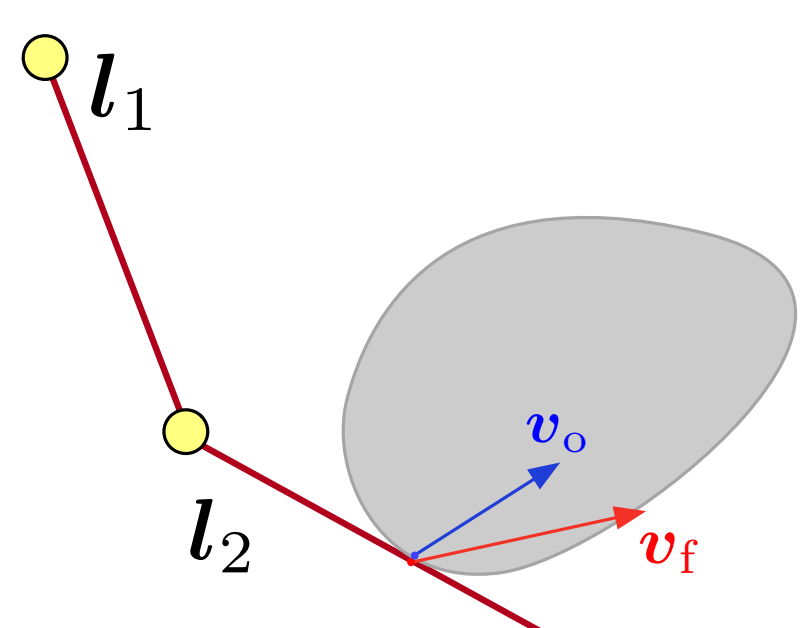
1. Contact Motions

- ❖ Simultaneous finger-object contacts
- ❖ Known object position and velocity

Object's location $c = c(\zeta)$

$$v = \dot{c}(\zeta, \dot{\zeta})$$

$$\dot{v} = \dot{v}(\zeta, \dot{\zeta}, \ddot{\zeta})$$



Contact modes:

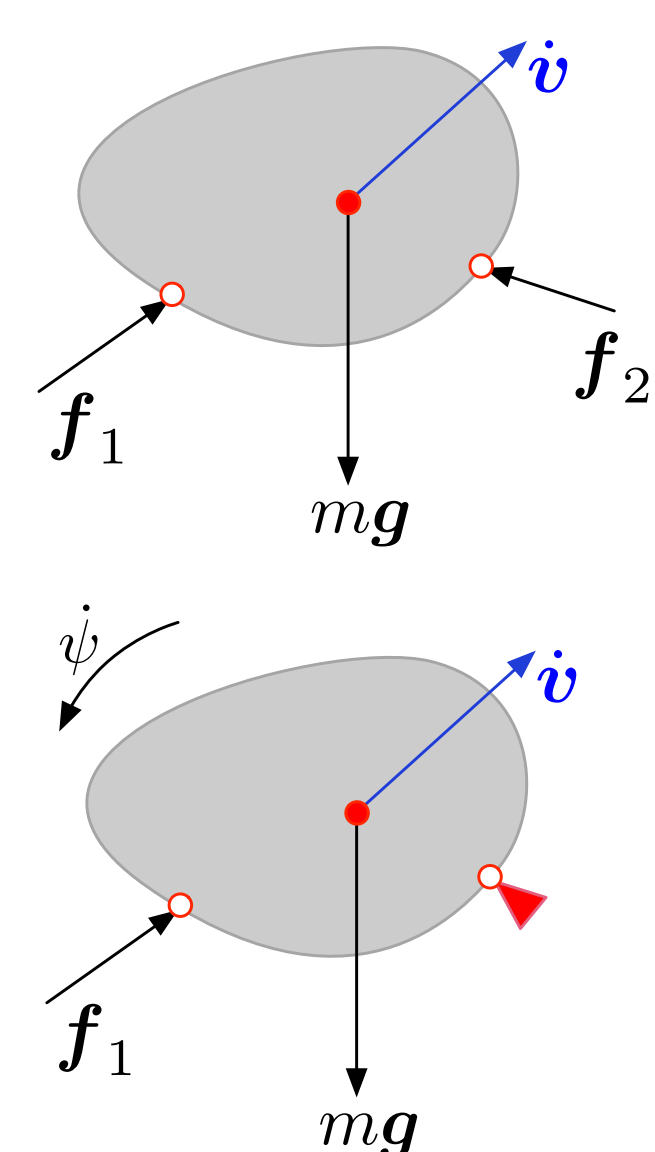
- ❖ *Sliding:*

$$v_{o\parallel} \neq v_{f\parallel}, \quad v_{o\perp} = v_{f\perp}$$

- ❖ *Rolling:*

$$v_o = v_f$$

2. Contact Mechanics



- ❖ *Both sliding:*
(Coulomb's law)

$$f_{1\parallel} = \mu f_{1\perp}$$

$$f_{2\parallel} = \mu f_{2\perp}$$

- ❖ *Rolling (e.g. on F2) + sliding:*
(Coulomb's law + rolling constraint)

$$f_{1\parallel} = \mu f_{1\perp}$$

$$\dot{\psi} = \dot{\psi}(v, \theta_2)$$

- ❖ *Both rolling:*
(over-constrained system: impossible)

$$f_i = B_i(\zeta)\ddot{\theta} + s_i(\zeta, \dot{\zeta}) + t_i(\zeta) \quad (i = 1, 2)$$

f_1 and f_2 in terms of the state of system $\zeta, \dot{\zeta}$ and $\ddot{\theta}$

Angular acceleration via Euler's equation:

$$\dot{\omega} = \frac{1}{I}(r_1 f_{1\parallel} + r_2 f_{2\parallel}) \quad (r_1, r_2 : \text{radius of curvature})$$

4. Joint Torque Control

$$\tau = \tau_{ff} + \tau_{fb}$$

- ❖ Feedforward component
(cancels all nonlinearities given position and velocity)

$$\tau_{ff} = \mathcal{C} + \mathcal{N}$$

- ❖ Feedback component
(provides correction torques to reduce errors using PD control law)

$$\tau_{fb} = \mathcal{M}(-K_v \dot{\theta}_e - K_p \theta_e)$$

Combine joint torques with finger dynamics:

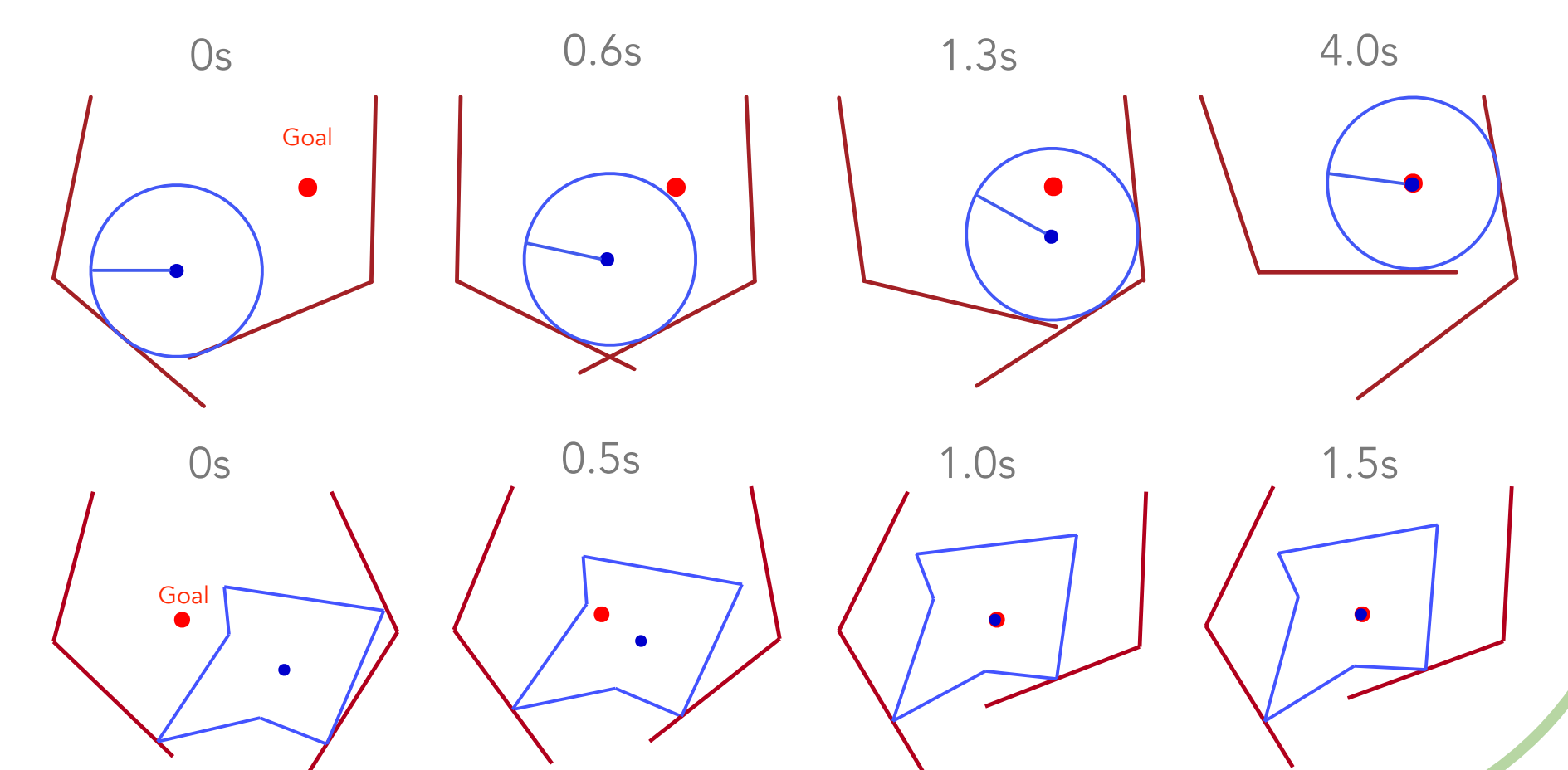
$$\tau = \mathcal{M}\ddot{\theta} + \mathcal{C} + \mathcal{N}$$

$$\tau = \mathcal{M}(-K_v \dot{\theta}_e - K_p \theta_e) + \mathcal{C} + \mathcal{N}$$

$$\ddot{\theta} = -K_v \dot{\theta}_e - K_p \theta_e$$

Exponential error decay

Finding proper gains K_v and K_p



3. Finger Dynamics

Lagrangian mechanics applied to a single finger with coupled joints.

$$\text{Lagrange-d'Alembert equation: } \frac{d}{dt} \frac{\partial \mathcal{T}}{\partial \dot{\theta}} - \frac{\partial \mathcal{T}}{\partial \theta} - \frac{\partial \mathcal{U}}{\partial \theta} = Q$$

$$\text{Generalized force: } Q = \tau + (-f) \cdot \frac{\partial p}{\partial \theta} = \tau - J^T f$$

Jacobian at contact point

$$\tau = M(\theta)\ddot{\theta} + C(\theta)\dot{\theta}^2 + N(\theta) + J^T f \quad (\text{single finger dynamics})$$

$$\tau = \mathcal{M}(\zeta)\ddot{\theta} + \mathcal{C}(\zeta, \dot{\theta}) + \mathcal{N}(\theta) \quad (\text{hand dynamics})$$

Linear relationship between motor torque τ and joint acceleration $\ddot{\theta}$.

